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7. Which one of the following gases has the lowest value of Henry's law constant ?

   a) N_2 b) He

   c) CO_2 d) H_2

8. P_1 and P_2 are the vapour pressures of pure liquid components, 1 and 2 respectively of an ideal binary solution if x_1 represents the mole fraction of component 1, the total pressure of the solution formed by 1 and 2 will be

   a) $P_1 + x_1 (P_2 - P_1)$

   b) $P_2 - x_1 (P_2 + P_1)$

   c) $P_1 - x_2 (P_1 - P_2)$

   d) $P_1 + x_2 (P_1 - P_2)$

9. Osometic pressure (π) of a solution is given by the relation

   a) $\pi = nRT$ b) $\pi V = nRT$

   c) $\pi RT = n$ d) none of these

10. Which one of the following binary liquid mixtures exhibits positive deviation from Raoults law ?

    a) Acetone + chloroform

    b) Water + nitric acid

    c) HCl + water

    d) ethanol + water

11. The Henry's law constants for two gases A and B are x and y respectively. The ratio of mole fractions of A to B is 0.2. The ratio of mole fraction of B and A dissolved in water will be

    a) $\frac{2x}{y}$ b) $\frac{y}{0.2x}$

    c) $\frac{0.2x}{y}$ d) $\frac{5x}{y}$

12. At 100°C the vapour pressure of a solution containing 6.5g a solute in 100g water is 732mm. If K_b = 0.52, the boiling point of this solution will be

    a) 102°C b) 100°C

    c) 101°C d) 100.52°C

13. According to Raoults law, the relative lowering of vapour pressure for a solution is equal to

    a) mole fraction of solvent

    b) mole fraction of solute

    c) number of moles of solute

    d) number of moles of solvent

14. At same temperature, which pair of the following solutions are isotonic ?

    a) 0.2 M $BaCl_2$ and 0.2M urea

    b) 0.1 M glucose and 0.2 M urea

    c) 0.1 M NaCl and 0.1 M K_2SO_4

    d) 0.1 M Ba $(NO_3)_2$ and 0.1 M Na_2 SO_4

15. The empirical formula of a non-electrolyte(X) is CH_2O. A solution containing six gram of X exerts the same osmotic pressure as that of 0.025M glucose solution at the same temperature. The molecular formula of X is

    a) $C_2H_4O_2$ b) $C_8H_{16}O_8$

    c) $C_4H_8O_4$ d) CH_2O

16. The K_H for the solution of oxygen dissolved in water is 4×10^4 atm at a given temperature. If the partial pressure of oxygen in air is 0.4 atm, the mole fraction of oxygen in solution is

    a) 4.6×10^3 b) 1.6×10^4

    c) 1×10^{-5} d) 1×10^5